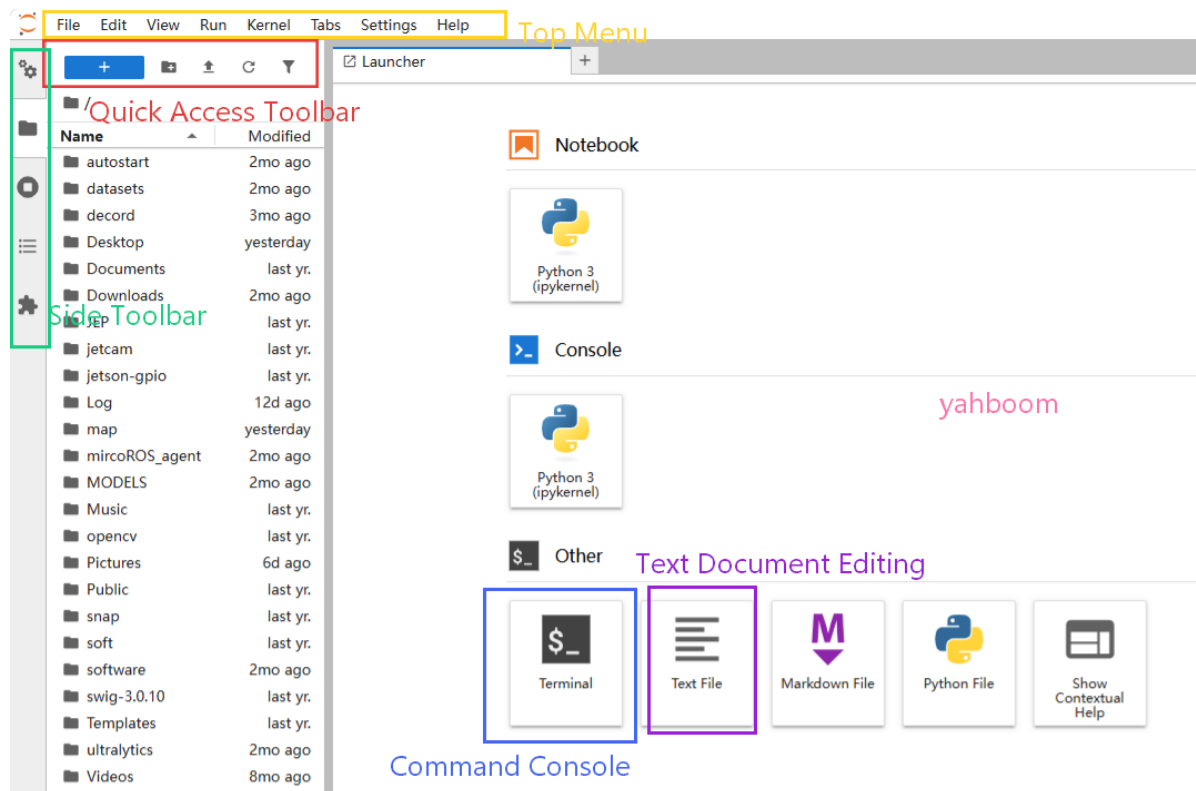


5. How to Use JupyterLab

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The JupyterLab interface serves as a dashboard, providing access to interactive iPython notebooks, the Raspblock folder structure, and a terminal window for accessing the Linux operating system. The initial view you encounter includes a **Menu Bar** at the top, a directory tree within the **Left Sidebar**, and a **Main Workspace** that initially opens to a "Launcher" page.

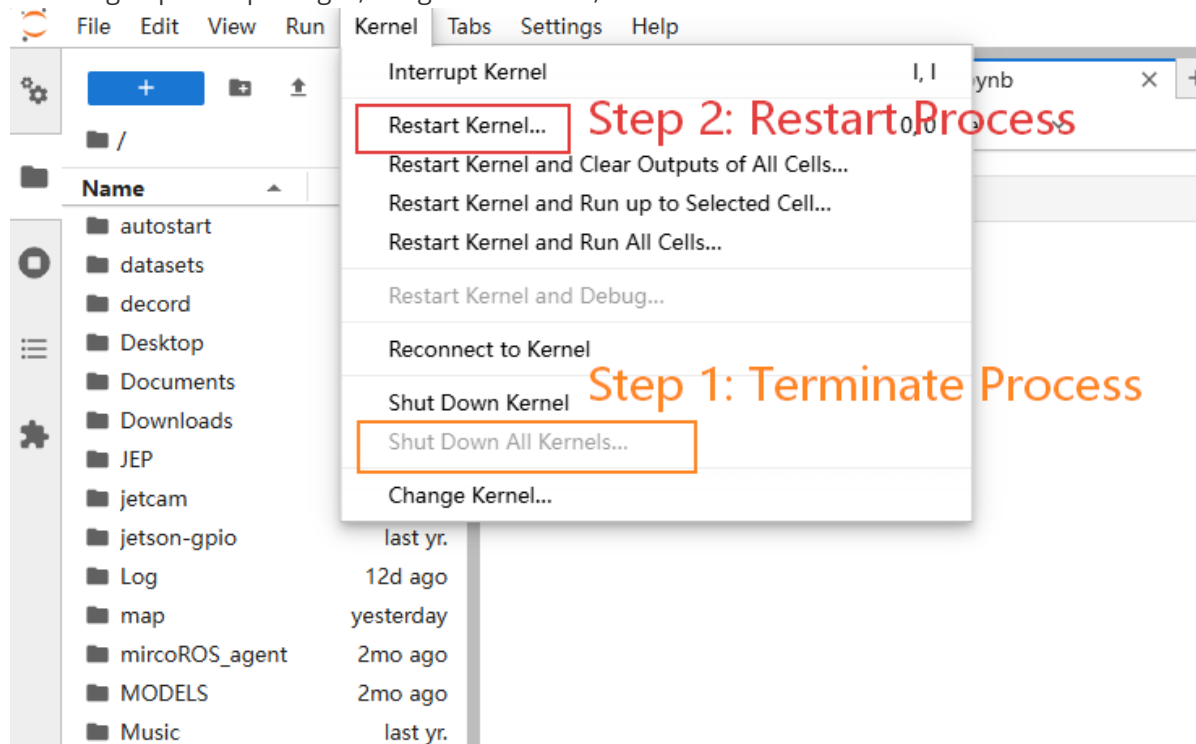


For full details regarding all features and menu operations, please refer to the [JupyterLab Interface](https://jupyterlab.readthedocs.io/en/stable/user/interface.html) documentation: <https://jupyterlab.readthedocs.io/en/stable/user/interface.html>

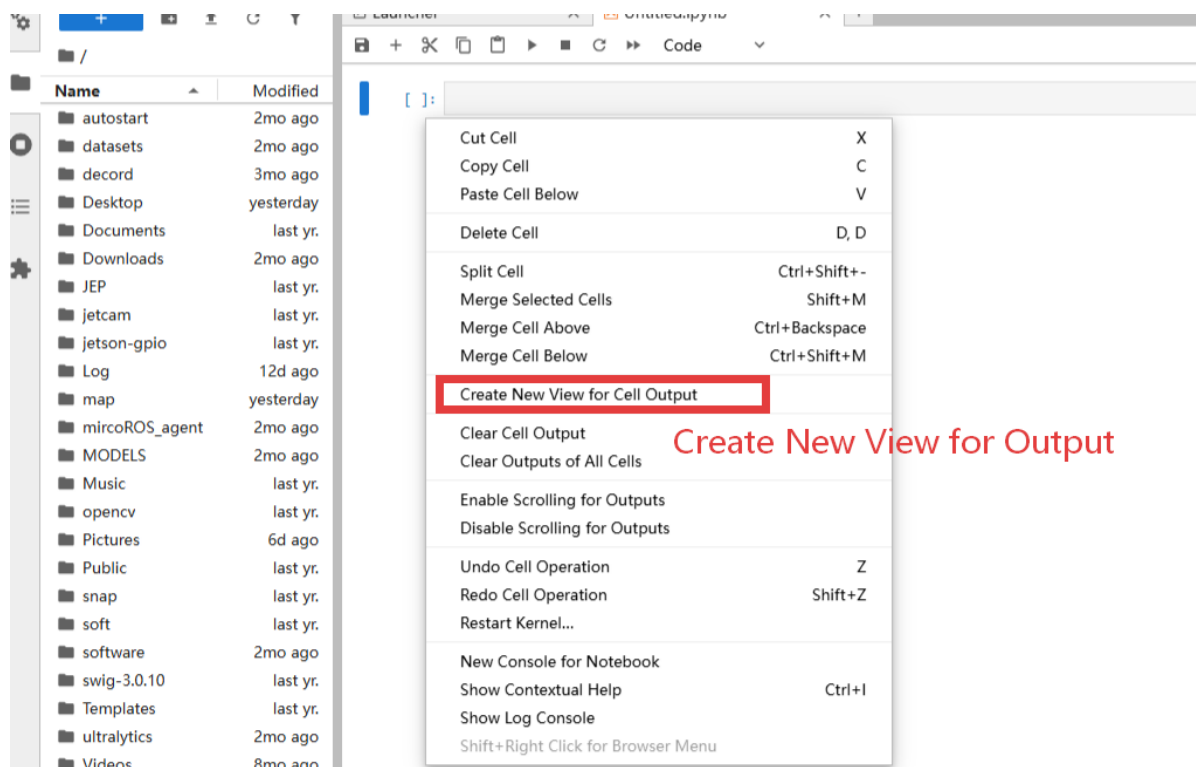
The following are some key features that are particularly useful in this course:

- **File Browser:** The File Browser located in the left sidebar allows you to navigate the file structure. Double-clicking a notebook or file will open it in the main workspace.
- **iPython Notebooks:** The interactive notebooks used in this course have the ".ipynb" file extension. When you double-click a notebook from the File Browser, it opens in the main workspace, and its execution process begins. Notebooks consist of text and code "cells." When a code cell is "run"—either by clicking the Run button at the top of the notebook or by using the keyboard shortcut [CTRL] + [ENTER]—the block of code within that cell is executed, and any output (if applicable) is displayed directly below the cell in the notebook. To the left of each executable cell, you will see an "execution count" or "prompt number" enclosed in brackets. If a cell takes more than a few seconds to run, you will see an asterisk (*) in that spot, indicating that the cell has not yet finished executing. Once the cell's processing is complete, a number will appear within the brackets.

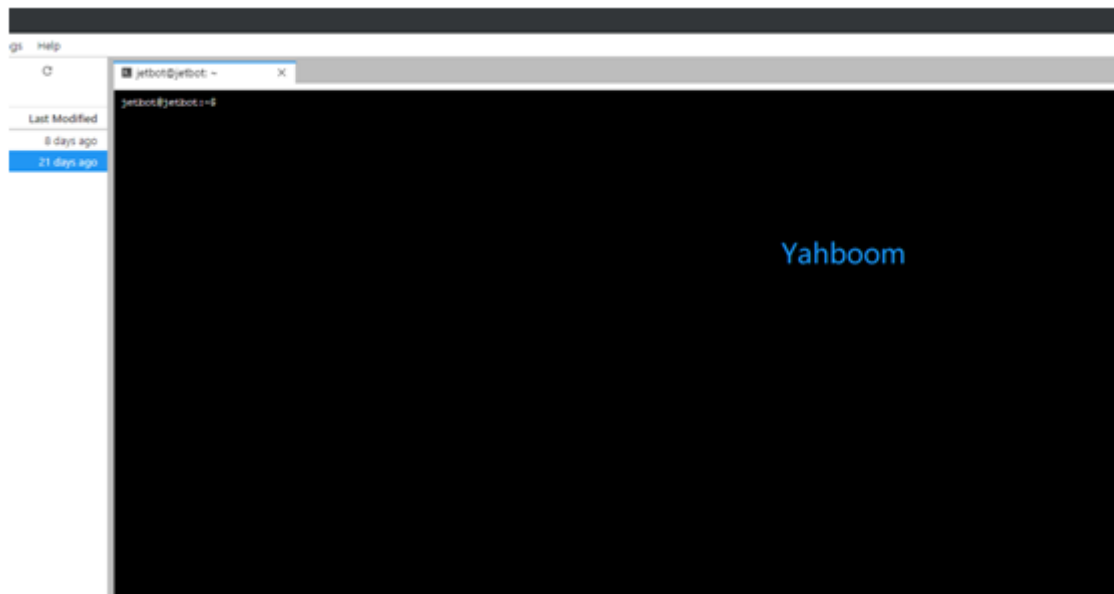
• **Kernel Operations:** The kernel for each running notebook is an independent process responsible for executing user code. When you open a notebook from the File Browser, the kernel starts automatically. The Kernel menu on the main menu bar contains commands to shut down or restart the kernel—operations you will need to perform periodically. Once a kernel is shut down, no code cells can be executed. When you restart a kernel, all data stored in memory—including imported packages, assigned variables, and so forth—is lost.



Cell Tabs: By right-clicking on a cell and selecting "Create New View for Output," you can move any cell into a new window tab within the main workspace. This allows you to continue scrolling down through your JupyterLab notebook while still keeping a specific cell—such as a camera view—in sight!



- **Terminal Window:** You can work directly within a [terminal window](#) on your Raspberry Pi via your remote Jupyter connection. On the Launcher page, click the "Terminal" icon located under the "Other" section. To bring up the Launcher page—if it is no longer visible—click the "+" icon at the top of the left sidebar.



Shortcuts are one of the greatest strengths of Jupyter Notebooks. Whenever you want to run a specific block of code, all you need to do is press `Ctrl + Enter`. Jupyter Notebooks offers a wide array of keyboard shortcuts that can help you save a significant amount of time.

Below, we have hand-picked a selection of shortcuts that will be immensely helpful as you get started. I highly recommend trying them out one by one as you read through this article. In the future, you won't be able to live without them!

Jupyter Notebooks provides two distinct keyboard input modes: Command Mode and Edit Mode. Command Mode binds keyboard inputs to notebook-level commands and is indicated by a gray cell border with a blue left margin. Edit Mode allows you to type text (or code) directly into the active cell and is indicated by a green cell border. You can toggle between Command Mode and Edit Mode by pressing `Esc` and `Enter`, respectively. Give it a try right now! Once you have entered Command Mode (at which point no cell is active), you can try the following shortcuts:

Shift+Enter: Run the current cell and select the next cell

Ctrl+Enter: Run the current cell

Alt+Enter: Run the current cell and insert a new cell below it

Y: Switch the cell to Code mode

M: Switch the cell to Markdown mode

A: Insert a new cell above the current one

B: Insert a new cell below the current one

X: Cut the selected cell

Shift+V: Paste the cell above the current one